# Sina Keshvadi

Dr. Sc. Sina Keshvadi
Assistant Professor
Department of Engineering
Faculty of Science
Thompson Rivers University
Email: skeshvadi@tru.ca

Sina Keshvadi
Thompson Rivers University
805 Tru Way
OM2812
Kamloops, BC, Canada
V2C 0C8

## Work Experience

Aug 2022 - Present
**Assistant Professor**
Engineering Dept., Faculty of Science
Thompson Rivers University, Kamloops, BC, Canada

Jan 2022 - Aug 2022
**Research Scientist**
Data-Driven Networking Protocols Team
Huawei Technologies Canada, Toronto, ON, Canada

Sept 2021 - Dec 2021
**Sessional Instructor**
Computer Science Dept., Faculty of Science
University of Calgary, Calgary, AB, Canada

Jul 2012 - Aug 2017
**Assistant Professor - Teaching Stream**
Computer Science and IT Dept.
Payame Noor University, Bushehr, Iran

## Education

2017 - 2021
**PhD in Computer Science**
University of Calgary, Calgary, Alberta, Canada
Advisor: Prof. Carey Williamson
Thesis: Traffic Characterization of Social Network Applications

2009 - 2011
**M.Eng.Sc. in Computer Engineering - Software**
Islamic Azad University, Ahvaz, Iran
Advisor: Prof. Amir Masoud Rahmani
Thesis: Design a Recommender System in a P2P File Sharing System

2005 - 2009
**B.S. in Computer Science**
Payame Noor University, Shiraz, Iran

# Publications

## Peer-Reviewed Journal Articles

*Student co-authors:* * undergraduate research student.

2026 Isaac Latta* and **Sina Keshvadi**, "Policy Without Enforcement? Web Security Posture of Canadian Public-Sector Websites," *Computers & Security (Elsevier)*, vol. 171, p. 105095, 2026.

2026 Nithiya Muruganandham*, Yogesh Sharma, and **Sina Keshvadi**, "An In-Depth Measurement of Security and Privacy Risks in the Free Live Sports Streaming Ecosystem," *Journal of Cybersecurity and Privacy*, vol. 6, no. 1, p. 8, 2026.

2026 Peyman Arebi and **Sina Keshvadi**, "TACO-BPNN: a hybrid ant colony optimization and backpropagation neural network approach for efficient controllability of temporal networks," *Evolutionary Intelligence (Springer)*, vol. 19, no. 1, p. 19, 2026.

2026 Damien Rossi*, **Sina Keshvadi**, and Yogesh Sharma, "ScionPathML: Enabling an Empirical Measurement Dataset and Benchmarks for Path-Aware Networking," *Telecom*, vol. 7, no. 4, p. 81, 2026.

2023 **Sina Keshvadi**, Shuihai Hu, Yi Lian*, and Geng Li, "OpenData: A Framework to Train and Deploy ML Solutions in Wide-Area Networks," *IEEE Network*, vol. 37, no. 5, pp. 25–32, 2023.

2022 **Sina Keshvadi** and Behnam Faghih, "Quality of Service Provisioning to Deliver Scalable High Efficiency Video Coding (SHVC) over SDN Networks," *International Robotics & Automation Journal*, vol. 8, no. 2, pp. 74–80, 2022.

## Peer-Reviewed Conference Publications

*Student co-authors:* * undergraduate research student.

2026 Isaac Latta* and **Sina Keshvadi**, "Evaluating the Security Posture of Canadian Government Authority Websites", In *2026 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*, Montreal, Canada, 2026.

2025 Matia Landry*, **Sina Keshvadi**, and Geoff Fink, "The Missing Software Engineering Course for Developing Essential Skills for Co-Op Success", In *2025 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*, Vancouver, Canada, 2025.

2025 Raiden Yamaoka*, Shaylee Broadfoot*, Catherine M. Tatarniuk, and **Sina Keshvadi**, "StudyHub: A Virtual Sanctuary for Focused Studying and Connection," In *Canadian Engineering Education Association (CEEA25)*, Montreal, Canada, 2025.

2025 Shaylee Broadfoot*, Raiden Yamaoka*, Catherine M. Tatarniuk, and **Sina Keshvadi**, "Developing an Online Cybersecurity Literacy Awareness Module for Post-Secondary Students," In *Canadian Engineering Education Association (CEEA25)*, Montreal, Canada, 2025.

2025 Nithiya Shri Muruganandham* and **Sina Keshvadi**, "Assessing the Network Security Threats Posed by Unregulated Free Live Streaming Services", In *2025 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*, Vancouver, Canada, 2025.

2024 **Sina Keshvadi**, Geoff Fink, and Catherine M. Tatarniuk, "The Development of an Extracurricular Course to Reduce the Knowledge Gap Between Industry and Academia in a Software Engineering Degree Program," In *Canadian Engineering Education Association (CEEA24)*, Edmonton, Canada, 2024.

2024 **Sina Keshvadi**, Leanne Mihalicz, and Mikhayla Maurer*, "Navigating the Software Engineering Landscape: A Career Mentorship Program for Academic-to-Industry Transition," In *Canadian Engineering Education Association (CEEA24)*, Edmonton, Canada, 2024.

2023 **Sina Keshvadi** and Yogesh Sharma, "Exploring HTTPS Certificate Ecosystem: Analyzing the Entire IPv4 Address Space", In *2023 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*, Regina, Canada, September 2023.

2023 **Sina Keshvadi** and Yogesh Sharma, "Enhancing Video Streaming Quality of Service with Software-Defined Networking and Network Slicing: A Scalable AV1 Approach", In *2023 IEEE 19th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)*, pp. 255-260, Montreal, Canada, June 2023.

2023 **Sina Keshvadi**, "Enhancing Western Organizational Cybersecurity Resilience through Tailored Education for Non-Technical Employees", In *2023 IEEE International Humanitarian Technology Conference (IHTC)*, Cartagena de Indias, Colombia, 2023.

2021 **Sina Keshvadi** and Carey Williamson, "An Empirical Measurement Study of Free Live Streaming Services", *Passive and Active Measurement: 22nd International Conference (PAM 2021)*, pp 111-127, Online Event, March 2021.

2020 **Sina Keshvadi**, Mehdi Karamollahi, and Carey Williamson, "Traffic Characterization of Instant Messaging Applications: A Campus-Level View", *IEEE 45th Conference on Local Computer Networks (LCN)*, pp. 225-232, Sydney, Australia, November 2020.

2020 **Sina Keshvadi** and Carey Williamson, "MoVIE: A Measurement Tool for Mobile Video Streaming on Smartphones", *Proceedings of the ACM/SPEC International Conference on Performance Engineering*, pp. 230-237, Edmonton, AB, Canada, April 2020.

2019 Steffen Berg Klenow, Carey Williamson, Martin Arlitt, and **Sina Keshvadi**, "Campus-level Instagram Traffic: A Case Study", *IEEE 27th International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS)*, pp. 228-234, Rennes, France, October 2019.

### Preprints and Manuscripts Under Review

2026 Lars Herschbach, Damien Rossi, and **Sina Keshvadi**, "Path Dynamics in a Deployed Path-Aware Network: A Measurement Study of SCIONLab," *Journal of Network and Systems Management*, revision submitted (major revision), 2026. Preprint: arxiv:2509.04695.

2026 **Sina Keshvadi**, "Performance Analysis of Dynamic Equilibria in Joint Path Selection and Congestion Control in Path-Aware Networks," Submitted to *Journal of Complex Networks*. Preprint: arXiv:2510.26060.

### Educational Resources

2026 **Sina Keshvadi**,
"Computer Network Security"
(textbook for the Computer Network Security course (CS 3261)),
TRU Open Learning.
Online version available at `https://keshvadi.github.io/security`.

## Teaching Experience

2022 - Present Assistant Professor, Department of Engineering, Thompson Rivers University
**Undergraduate**
*Software Security* (W23, W24, W25, W26)
*Web App Development* (W23, W24, W25, W26)
*Practical Cloud Computing* (F23, F24, F25, F26)
*Software Requirement and Spec* (F22, F23)
**Graduate**
*Applied AI Agents* (F26)

2020 - 2021 Sessional Lecturer, Computer Science Dept., University of Calgary
*CPSC 441 (Computer Networks)* (F20, F21; ~125 students per section)

2017 - 2021 Teaching Assistant, Computer Science Dept., University of Calgary

*CPSC 447 (Operating Systems)* (F17, W18, F18)
*CPSC 441 (Computer Networks)* (W19)

2012 - 2017 Assistant Teaching Professor, Payame Noor University, Computer Science and IT Dept., Iran
*Principles of Operating Systems* (F12, F13, F15, F16)
*Computer Networks* (W12, W13, W16)
*Data Structures and Algorithms* (W12, W13, W16)
*Web Programming* (W15, W16)
*Programming*(W12, W13, W14)
*Compiler Construction* (W15)
*Computer Simulation* (W12)

## Research Grants

2026 Mitacs Globalink Research Internship, $12,000
(Estimated value; paid directly to student)
Role: Supervisor
Student Supported: Natasha Zaheer (Pakistan)

2026 Mitacs Globalink Research Internship, $12,000
(Estimated value; paid directly to student)
Role: Supervisor
Student Supported: Faizan Anwar (Pakistan)

2026 Mitacs Globalink Research Internship, $12,000
(Estimated value; paid directly to student)
Role: Supervisor
Student Supported: Emilio Moreno Almanza (Mexico)

2025 CSA Research Scholarships.
CSA Group, $6,000
Role: Supervisor
Student Supported: Isaac Latta (undergraduate engineering)

2025 SoTL Scholars Development Program, $2,500
Centre for Excellence in Learning & Teaching, Thompson Rivers University

2025 Internal Research Fund (IRF) Grant, $5,000
Thompson Rivers University

2025 Mitacs Globalink Research Internship, $12,000
(Estimated value; paid directly to student)
Role: Supervisor
Student Supported: Alissa Baumeister (Germany)

2025 Mitacs Globalink Research Internship, $12,000
(Estimated value; paid directly to student)
Role: Supervisor
Student Supported: Damien Rossi (France)

2025 Mitacs Globalink Research Internship, $12,000
(Estimated value; paid directly to student)
Role: Supervisor
Student Supported: Lars Herschbach (Germany)

2024 Undergraduate Research Apprenticeship Fund, $3,000
Thompson Rivers University
UG Student: Michael Nolander

2024 Undergraduate Research Apprenticeship Fund (Co-Investigator), $3,000
Thompson Rivers University
Co-investigator: Dr. Catherine Tatarniuk
UG Student: Shaylee Broadfoot

2024 Undergraduate Research Apprenticeship Fund (Co-Investigator), $3,000
Thompson Rivers University
Co-investigator: Dr. Catherine Tatarniuk
UG Student: Raiden Yamaoka

2024 Google Cloud Research Credits, $7,021 (USD Cloud Credits)
Google

2024 Mitacs Globalink Research Internship, $12,000
(Estimated value; paid directly to student)
Role: Supervisor
Student Supported: Nithiya Shri Muruganandham (India)

2024 Mitacs Globalink Research Internship, $12,000
(Estimated value; paid directly to student)
Role: Supervisor
Student Supported: Varun Mayilvaganan (India)

2023 TRU Faculty Research Funds (non-competitive), $7,500
Thompson Rivers University

2023 TRU Internal Research Fund (IRF), $4,824
Thompson Rivers University

### Teaching Grants

2026 Google Cloud credits for teaching, USD $2,000
In-kind credits for students ($50 per student) for course projects and labs.
Role: Instructor.

2026 Google Skills / Google Cloud lab credits, USD $5,000
In-kind credits for students to complete Google labs in course work.
Role: Instructor.

### Scholarships

2020 Computer Science Alumni Chapter Graduate Scholarship (competitive)
University of Calgary

2019 Computer Science Alumni Chapter Graduate Scholarship (competitive)
University of Calgary

2019 International Graduate Recruitment Award (non-competitive)
University of Calgary

## Professional Service

### Graduate Examinations and Committee Member

2026 Panelist, Final Project Presentations (Master's)
Students: Mohammed Danial Shaikh, Onyejiaka Emmanuella Obumneke, Haoran He, Arpitha Thippeswamy

2025 Chair of Defence Committee, (Master's defense)
Student: Jamil Reza

2025 Judge, Engineering Capstone Final Project Presentations

2024 Panelist, Final Project Presentations (Master's)
Students: Jiaze Wang, Zijian Wang, Max Li, Lei Meng

2024 Moderator, Final Project Presentations (Master's)
Students: Qiuhan Li, Yueyue Zhang

2024 Chair of Defence Committee (Master's defense)
Student: Komeil Sharafi

2024 Judge, Engineering Capstone Final Project Presentations

2023 Chair of Defence Committee (Master's defense)
Student: Bindubritta Acharjee

### Internal Service

*Thompson Rivers University, Canada*

2023 - Present Committee Member for Academic Planning and Priorities Committee (APPC)
Thompson Rivers University

2024 - Present Committee Member for Science Curriculum Review Committee (SCRC)
Faculty of Science

2022 - Present Committee Member for Unspent Professional Allowance Allocation Committee
Faculty of Science

2024 - Present Chair of Engineering Design and Exams Committee
Department of Engineering

2023 - Present Committee Member for Health & Safety Committee
Department of Engineering

2022 - Present Committee Member for Course and Curriculum Planning Committee
Department of Engineering

2022 - Present Committee Member for Department Recruitment & Retention Committee
Department of Engineering

*University of Calgary, Canada*

2020 - 2021 Graduate Student Representative for Teaching & Learning Committee, Faculty of Science

### External Service

2024-2026 Reviewer: The Journal of Supercomputing
2026 Reviewer: Journal of King Saud University Computer and Information Sciences
2025-2026 Reviewer: Canadian Engineering Education Association Conference
2025 Reviewer: Thompson-Okanagan Teaching & Learning Conference
2024 - Present Researcher Member, ACCP Alliance Canada
2024 - Present Reviewer: Canadian Journal of Learning and Technology
2024 Reviewer: The Journal of Cluster Computing
2023 - Present AWS Academy Educator at TRU
2023 - Present Reviewer: Journal of Complex Networks
2023 - Present Reviewer: International Journal of Mobile Network Design and Innovation
2022 - Present Reviewer: IEEE Canadian Journal of Electrical and Computer Engineering
2022 Program Committees: Passive and Active Measurement Conference, 2023
2019 - 2020 Reviewer: International Journal of Sensors, Wireless Communications and Control

## Undergrad Supervision

*Thompson Rivers University, Canada*

2026 **Emilio Moreno Almanza** — *Mitacs Globalink Research Internship Program*
*Project:* State-Aware Multipath Scheduling for MPQUIC over SCION

2026 **Faizan Anwar** — *Mitacs Globalink Research Internship Program*
*Project:* Stability Modeling for High-Path-Count Congestion Control

2026 **Natasha Zaheer** — *Mitacs Globalink Research Internship Program*
*Project:* Empirical Evaluation of MPQUIC Performance on the Global SCION Testbed

2025 - 2026 **Esteban Flores Calderon** — *SoTL Grant*
*Project:* Cloud Computing and Professional Identity

2025 **Alissa Baumeister** — *Mitacs Globalink Research Internship Program*
*Project:* Path-aware network simulator

2025 **Lars Herschbach** — *Mitacs Globalink Research Internship Program*
*Project:* Large-scale SCIONLab measurement

2025 **Damien Rossi** — *Mitacs Globalink Research Internship Program*
*Project:* ML benchmarking / ScionPathML toolkit

2024 - 2025 **Raiden Yamaoka** — *TRU URA*
*Project:* Full-stack "StudyHub" application
*Co-supervisor:* Dr. Catherine Tatarniuk

2024 - 2025 **Shaylee Broadfoot** — *TRU URA*
*Project:* Interactive phishing-detection training module
*Co-supervisor:* Dr. Catherine Tatarniuk

2024 - 2025 **Michael Nolander** — *TRU URA*
*Project:* ML benchmarking for phishing URL detection

2024 **Matia Landry** — *TRU Work-Study*
*Project:* Open-access webbook on industry tools

2024 **Nithiya Muruganandham** — *Mitacs Globalink Research Internship Program*
*Project:* Privacy/security risks in free live streaming

2024 **Varun Mayilvaganan** — *Mitacs Globalink Research Internship Program*
*Project:* GAN-resistant phishing detection

2024 **Megan Rouault** — *TRU IRF*
*Project:* Campus cybersecurity awareness study

2024 **Trevor Drayton** — *TRU IRF*
*Project:* Campus cybersecurity awareness study

2023 **Chris Coulthard** — *TRU IRF*
*Project:* AWS performance monitoring + ML anomaly detection

2023 **Mikhayla Maurer** — *TRU SoTL*
*Project:* Full-stack mentoring platform
*Co-supervisor:* Dr. Leanne Mihalicz

2023 **Andrew Peaker, Nicholas Plumtree, and Chase Swanson** — *TRU ACERS Competition*
*Project:* Wi-Fi subsystem for autonomous vehicle (First-place team)